Towards Tensorial Strain Reconstruction from Eddy-Current Testing in Laser Powder Bed Fusion

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Abstract

Eddy-current testing (ECT) is an attractive in-situ probe for laser powder bed fusion (LPBF) because it is non-contact, fast, and sensitive to the electrical conductivity, which depends on mechanical strain. Prior work established a *scalar* relation $\Delta\sigma/\sigma_0 = \kappa\,\varepsilon$ between the relative conductivity change and the strain, calibrated to $\kappa = -0.326$ for 316L stainless steel. However, the residual strain in an as-built part is a second-order *tensor*, whereas a coil measures a single scalar impedance: the reconstruction of a six-component tensor from scalar data is fundamentally under-determined. We present a framework that resolves this tension by combining (i) a tensorial elastoresistivity model that generalises κ to a fourth-order coupling tensor; (ii) a finite-element (FEM) reciprocity analysis that quantifies which conductivity components a given probe senses; (iii) an optimal-design procedure for directional-probe sets (a conductivity "rosette"); and (iv) a regularised, prior-informed inversion that fuses the measurements with a thermomechanical process simulation (data assimilation). Along the way, cross-validating the analytical model against the FEM — whose impedance we verify against an independent ohmic-loss balance to better than 10^{-5} — exposed two implementation pitfalls in our realisation of the classical Dodd–Deeds model: a reflection-coefficient sign and a units error in the propagation parameter. Such forward-model bugs survive a self-consistent synthetic round trip (an inverse crime) but are caught by a dissimilar redundant model; after correction the analytical and FEM normalised impedances agree to 0.9% in resistance and 2.8% in reactance. On synthetic LPBF builds, a single normal coil resolves one strain degree of freedom, an in-plane rosette resolves three, and a six-direction tilted set resolves all six; the prior supplies the components that the probes cannot observe. The complete workflow is implemented in an open pipeline that outputs a full strain-tensor field.

1 Introduction

Residual stress and strain are central concerns in metal additive manufacturing: they drive distortion, cracking, and premature failure. In-situ monitoring that could map the evolving strain field during a build would enable closed-loop process control. Eddy-current testing (ECT) is a promising modality [5]: a coil driven at frequency ω induces eddy currents in the conductive part, and the resulting change in coil impedance encodes the local electrical conductivity σ. Because σ depends on strain through the elastoresistive effect, ECT indirectly senses strain.

The thesis underlying this work [15] established two pillars: (a) reconstruction of as-built geometry from ECT data for immersed-boundary (finite cell) simulation, and (b) a calibrated *scalar* law

$$\frac{\Delta\sigma}{\sigma_0} = \kappa\,\varepsilon, \qquad \kappa = -0.326 \;\; (316\text{L}), \tag{1}$$

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with coefficient of determination $R^2 = 0.992$ in uniaxial tension. Equation (1) is, however, a one-dimensional projection of a richer physical reality. In three dimensions the strain ε is a symmetric second-order tensor with six independent components, while a coil reports a single scalar. Recovering six numbers from one is impossible without additional information.

Related work. That conductivity depends on elastic strain through the piezoresistive (elastoresistive) effect underpins an established line of eddy-current residual-stress assessment developed largely by the Nagy group: high-frequency apparent-conductivity spectroscopy and iterative inversion recover near-surface residual-stress depth profiles in shot-peened superalloys [6, 7, 8]. That work also established a key confounder — cold work and dislocation-density changes alter conductivity alongside elastic strain — which is especially pronounced in LPBF microstructures and which we revisit in Section 8. In metal additive manufacturing, recoater-mounted ECT is already used for layer-wise density and defect monitoring [9, 11], with part-temperature compensation shown to be essential [10]; residual stress is a central concern of the field [12, 13] but is not yet mapped in situ. Our contribution is therefore not "ECT senses strain" but a *tensorial* observability framework, optimal rosette design, and simulation-prior fusion that together target the strain *tensor* in LPBF.

This paper develops the methodology and tooling to close that gap. We make four contributions: a tensorial elastoresistivity model (Section 3); an FEM-based observability analysis (Section 4) built on a forward model that we verify and cross-validate (Section 2); an optimal directional-probe ("rosette") design (Section 5); and a prior-informed tensor inversion realised inside an integrated pipeline (Sections 6–7).

2 Forward model and validation

2.1 Analytical Dodd–Deeds model

For a cylindrical multi-turn coil of inner/outer radius r_i, r_o, length ℓ, n turns and lift-off s above a conductive half-space, the impedance follows the Dodd–Deeds formulation [1, 2, 4]

$$Z = \frac{\mathrm{j}\omega\mu_0\pi n^2}{\ell^2(r_o - r_i)^2}\int_0^\infty \frac{J^2(\alpha r_i, \alpha r_o)}{\alpha^5}\Big[2\ell + \tfrac{1}{\alpha}\big(2e^{-\alpha\ell} - 2 + P(\alpha)\,\Gamma(\alpha)\big)\Big]\,\mathrm{d}\alpha, \tag{2}$$

where $P(\alpha) = e^{-2\alpha(\ell+s)} + e^{-2\alpha s} - 2e^{-\alpha(\ell+2s)}$, $\gamma = \sqrt{\alpha^2 + \mathrm{j}\omega\mu\sigma}$ and Γ is the half-space reflection coefficient; $J(\alpha r_i, \alpha r_o) = \int_{\alpha r_i}^{\alpha r_o} x\,J_1(x)\,\mathrm{d}x$ is the coil's Bessel kernel and $\mu = \mu_0$ for the non-magnetic 316L considered here. The normalised impedance is $Z_{\mathrm{norm}} = (Z_0 - Z)/\,\mathrm{Im}(Z_0)$ with Z_0 the free-space (self) impedance.

The physically correct reflection coefficient for a non-magnetic half-space is

$$\Gamma(\alpha) = \frac{\alpha - \gamma}{\alpha + \gamma}, \tag{3}$$

which tends to -1 as $\sigma \to \infty$: eddy currents then reduce the net inductance and add resistance, as required physically. A realisation that uses the opposite sign, $(\gamma - \alpha)/(\alpha + \gamma)$, predicts the unphysical opposite (increasing reactance, negative resistance over a passive conductor). This pitfall is invisible to a self-consistent synthetic round trip — an *inverse crime* — but corrupts the inversion of real data; we caught it only by cross-validating against the dissimilar FEM model below.

2.2 Axisymmetric finite-element model

To handle geometries where the half-space assumption fails (finite parts, layers, defects), we solve the time-harmonic eddy-current problem for the azimuthal magnetic vector potential A_ϕ

Table 1: FEM forward-model results and internal-consistency (verification) checks (reference coil over a generic conductor, $\sigma = 16.2$ MS/m, 240 kHz; realistic 316L values are used in Section 8).

Quantity	Value	Check
Z_0 (coil in air)	$0 + 5.18 \times 10^4\,\mathrm{j}\,\Omega$	purely inductive
Z (over conductor)	$622 + 3.73 \times 10^4\,\mathrm{j}\,\Omega$	$\mathrm{Re} > 0$, $\mathrm{Im} < \mathrm{Im}\,Z_0$
$\mathrm{Re}(Z)$ flux linkage	$622.35\,\Omega$	–
R from ohmic loss	$622.35\,\Omega$	agree to $< 10^{-5}$
mesh convergence	order 3 $\approx$ order 4	5 significant figures

in axisymmetric coordinates:

$$\frac{1}{\mu}\int\left[\nabla A_\phi\cdot\nabla v + \frac{A_\phi v}{\rho^2}\right]\rho\,\mathrm{d}\rho\,\mathrm{d}z + \mathrm{j}\omega\int \sigma A_\phi v\,\rho\,\mathrm{d}\rho\,\mathrm{d}z = \int J_s\,v\,\rho\,\mathrm{d}\rho\,\mathrm{d}z, \tag{4}$$

with $A_\phi = 0$ on the symmetry axis and the truncated outer boundary. The coil impedance is obtained from the flux linkage of the winding,

$$\Lambda = \frac{n}{S_{\text{coil}}}\iint_{\text{coil}} 2\pi\rho\,A_\phi\,\mathrm{d}\rho\,\mathrm{d}z, \qquad Z = \mathrm{j}\omega\Lambda/I, \tag{5}$$

and Z_0 is computed by the same model with $\sigma = 0$, giving a self-consistent normalisation.

The FEM is mesh-converged and internally consistent: the resistance extracted from the flux-linkage impedance (5) equals a direct volume integral of the ohmic loss $\frac{1}{2}\int \sigma|E|^2\,\mathrm{d}V$ to better than 10^{-5} (Table 1) — a *verification* check, distinct from experimental validation. Cross-validating the analytical model against this FEM exposed a second implementation pitfall in our realisation: a units error in the propagation parameter γ (a missing factor of μ_0, so that $\omega\mu\sigma$ no longer had units of m^{-2}), which made the conductor appear nearly perfect ($\Gamma \to -1$) — an essentially correct reactance but a loss suppressed by nearly three orders of magnitude. With both the reflection sign (3) and the units corrected, the analytical and FEM *normalised* impedances agree to 0.9% in resistance and 2.8% in reactance (Fig. 1); the small reactance residual is the finite FEM truncation versus the analytical half-space. The analytical integral (2) is converged for $k_{\max} \sim 6000$, within a stable plateau before the oscillatory Bessel kernel degrades the quadrature. This benchmark uses a reference coil over a generic conductor; realistic 316L parameters and their resolution implications are treated in Section 8.

3 From scalar to tensor: elastoresistivity

Strain breaks the isotropy of the material, so under load the conductivity is a second-order tensor σ_{ij} that couples to strain through a fourth-order *elastoresistivity* tensor K:

$$\frac{\Delta\sigma_{ij}}{\sigma_0} = K_{ijkl}\,\varepsilon_{kl}. \tag{6}$$

For an isotropic base material K has the isotropic form $K_{ijkl} = a\,\delta_{ij}\delta_{kl} + b(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$, i.e. two independent constants. Parameterising them as a longitudinal $\kappa_\parallel$ and a transverse $\kappa_\perp$ coefficient,

$$\frac{\Delta\sigma_{ii}}{\sigma_0} = \kappa_\parallel\varepsilon_{ii} + \kappa_\perp\sum_{k\neq i}\varepsilon_{kk}, \qquad \frac{\Delta\sigma_{ij}}{\sigma_0} = (\kappa_\parallel - \kappa_\perp)\varepsilon_{ij} \ \ (i \neq j). \tag{7}$$

The scalar law (1) is the uniaxial projection of (6): the relative conductivity change sensed along a unit direction $\hat{\boldsymbol{n}}$ in a uniaxial-stress test is $\hat{\boldsymbol{n}}^\top(\Delta\sigma/\sigma_0)\hat{\boldsymbol{n}}$, which varies strongly with

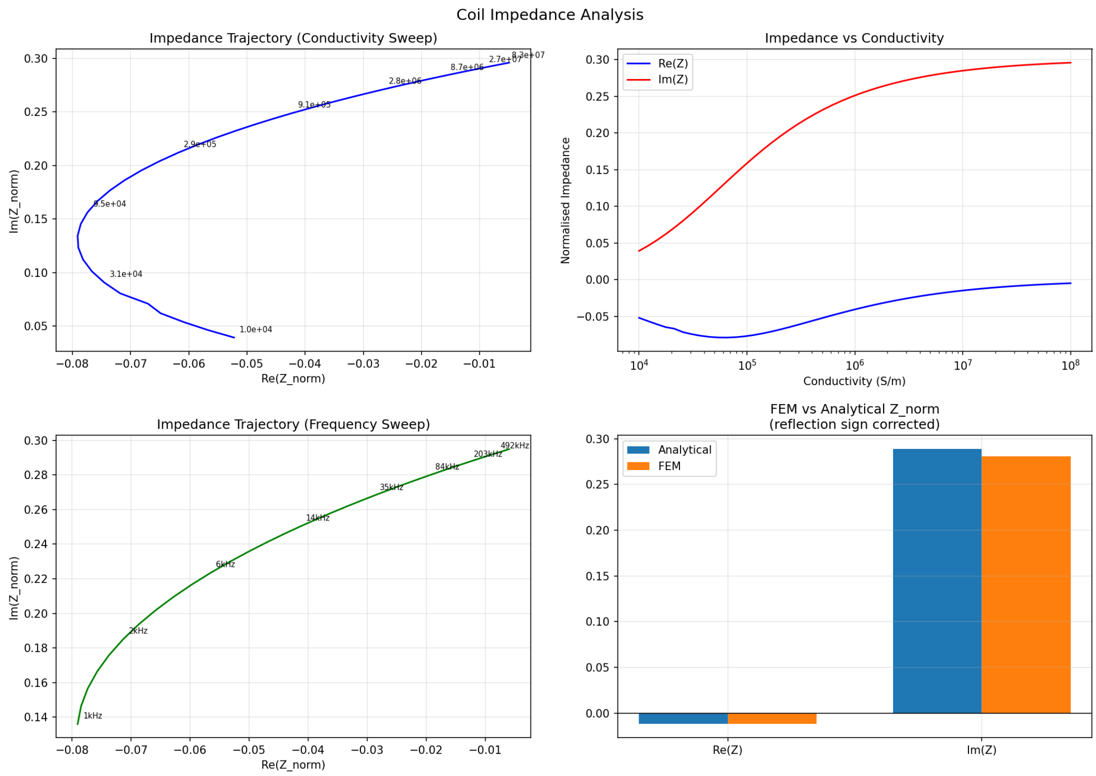

Figure 1: Coil impedance analysis. Analytical conductivity- and frequency-sweep trajectories (top, bottom-left) and the FEM-vs-analytical normalised impedance (bottom-right). After correcting the reflection sign and a units error in the propagation parameter, the FEM and analytical normalised impedances agree in both resistance (0.9%) and reactance (2.8%).

$\hat{\boldsymbol{n}}$ (Fig. 2). The measured value -0.326 is therefore one directional projection, not a material invariant; separating $\kappa_{\|}$ and $\kappa_{\perp}$ requires loading along several orientations. A purely volumetric response ($\kappa_{\|} = \kappa_{\perp}$) reduces K to rank one, so conductivity then constrains only the volumetric strain $\operatorname{tr}\boldsymbol{\varepsilon}$.

Constants and provenance. The elastoresistivity constants are not yet independently calibrated; we anchor them to the available data. A single uniaxial test fixes only one projection, so we choose $\kappa_{\|}$ such that the along-load uniaxial projection reproduces the thesis scalar ($\kappa_{\|} - 2\nu\kappa_{\perp} = -0.326$ at $\nu = 0.30$), giving $\kappa_{\|} = -0.374$ for an illustrative $\kappa_{\perp} = -0.08$. The result figures use the round value $\kappa_{\|} = -0.40$ (along-load projection -0.352, within 8% of -0.326). Separating $\kappa_{\|}$ from $\kappa_{\perp}$ requires multi-orientation loading (Section 8). Crucially, the observability results depend on these constants only through the ratio $\kappa_{\perp}/\kappa_{\|}$, shown to be robust across its plausible range in Section 5.

4 Observability analysis

4.1 Reciprocity sensitivity kernel

By driving-point reciprocity [3], a small conductivity perturbation changes the coil impedance by

$$\delta Z = -\frac{1}{I^2} \int \delta\sigma\, \boldsymbol{E} \cdot \boldsymbol{E}\, \mathrm{d}V. \tag{8}$$

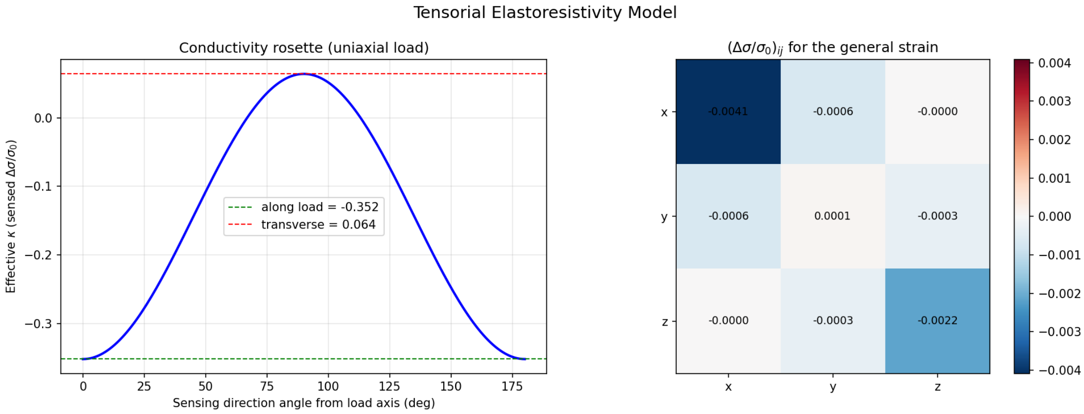

Figure 2: Tensorial elastoresistivity. Left: the "conductivity rosette" — the effective κ sensed in a uniaxial test depends on the sensing angle, swinging from -0.35 (along load) to $+0.06$ (transverse). Right: the induced $(\Delta\sigma/\sigma_0)_{ij}$ tensor for the strain $\varepsilon_{xx} = 1.0\%$, $\varepsilon_{yy} = -0.3\%$, $\varepsilon_{zz} = 0.4\%$, $\varepsilon_{xy} = 0.2\%$, $\varepsilon_{yz} = 0.1\%$ (illustrative $\kappa_{\|} = -0.40$, $\kappa_{\perp} = -0.08$).

For the axisymmetric coil the field is purely azimuthal, $\boldsymbol{E} = -\mathrm{j}\omega A_\phi\,\hat{\boldsymbol{\phi}}$, so $\boldsymbol{E}\cdot\boldsymbol{E} = E_\phi^2$ and only the local azimuthal (in-plane) component of a tensor perturbation is sensed. We confirmed (8) against finite-difference re-solves on a fixed mesh, with the ratio approaching unity as the perturbation shrinks (0.9963 at 0.5%). Consequently a normal coil senses only in-plane conductivity and is structurally *blind* to σ_{zz}; its sensitivity is concentrated within a skin depth of the surface (Fig. 3, top-left).

4.2 Directional probes and the conductivity rosette

A directionally selective probe senses, to leading order, the effective conductivity along its sensing direction,

$$y(\hat{\boldsymbol{n}}) = \hat{\boldsymbol{n}}^\top \boldsymbol{\sigma}\,\hat{\boldsymbol{n}}, \tag{9}$$

the conductivity analogue of a mechanical strain-gauge rosette.

Equation (9) is a leading-order model: it assumes each probe drives current cleanly along $\hat{\boldsymbol{n}}$. Accessed from the build surface this is well justified in-plane but optimistic out-of-plane, because the induced eddy currents are confined within a skin depth and flow essentially parallel to the surface — the axisymmetric solve of Section 2 makes this exact, with $\boldsymbol{E}$ purely azimuthal so that a normal coil is rigorously blind to σ_{zz}. A coil merely tilted above the surface therefore gains only limited σ_{zz} sensitivity; genuine out-of-plane access calls for edge/side probing, elongated or rotating-field probes, or multi-frequency depth profiling. We accordingly treat the six-direction, full-rank result below as an *upper bound* on what directional diversity can deliver, and take the in-plane rosette combined with the simulation prior (Section 6) as the practical configuration; extracting the true out-of-plane row of M from a 3-D anisotropic eddy-current model is left to future work.

Stacking M probes gives a linear map $\boldsymbol{y} = \mathsf{M}\,\boldsymbol{\sigma}_{\mathrm{v}}$ from the conductivity Voigt vector to the measurements; composing with K yields the strain map $\boldsymbol{y} = \mathsf{A}\,\boldsymbol{\varepsilon}_{\mathrm{v}}$ with $\mathsf{A} = \sigma_0\,\mathsf{M}\,\mathsf{K}$. The singular value decomposition of A gives the number of resolvable strain combinations (rank), their conditioning, and the null space (blind directions). Table 2 summarises the resolving power of representative configurations.

Table 2: Resolvable strain degrees of freedom and conditioning by probe set. The tilted six-probe row presumes idealised out-of-plane access and is an upper bound (Section 4).

Configuration	resolvable DOF (of 6)	condition number
Single normal coil	1	1.0
In-plane rosette (0/45/90)	3	2.6
In-plane rosette (0/60/120)	3	2.2
Optimised in-plane rosette	3	1.9
Rosette + tilted probes (6)	6	2.5

5 Probe-configuration design

Given K and the model (9), we choose sensing directions that make the inversion full-rank and well-conditioned by minimising the condition number of A subject to a rank constraint. For three in-plane probes the optimiser improves on the naive 0/45/90 (cond = 2.57) and the delta rosette 0/60/120 (2.18), reaching 1.93. A six-direction set that includes tilted probes attains full rank (6) at cond = 2.50 (Fig. 4) — an upper bound that presumes the idealised out-of-plane access discussed in Section 4. The ceiling is set by rank K: directional diversity helps only insofar as the elastoresistive response is anisotropic. Excitation frequency provides a complementary, depth-selective degree of freedom through the skin depth $\delta = \sqrt{2/(\omega\mu\sigma)}$ (Fig. 4, bottom-right).

The ceiling rank K is itself set by the material. Figure 5 sweeps the elastoresistivity ratio $\kappa_\perp/\kappa_\parallel$: the resolvable DOF stay full for any anisotropic response, but the conditioning of the strain map degrades smoothly toward the volumetric limit $\kappa_\perp \to \kappa_\parallel$, where the rank finally collapses to one. At the thesis-anchored ratio (≈ 0.21) both the in-plane rosette and the full set are comfortably conditioned, so the design is insensitive to the under-determined split between $\kappa_\parallel$ and $\kappa_\perp$.

6 Tensor-strain inversion with a simulation prior

Because realistic probe sets are often rank-deficient (e.g. surface access precludes observing σ_{zz}), we estimate the strain by maximum a posteriori (MAP) inversion [16] that fuses measurements with a process-simulation prior $\boldsymbol{\varepsilon}_{\text{prior}}$ (data assimilation). With Gaussian measurement noise (σ_d) and prior (σ_p),

$$\boldsymbol{\varepsilon}^\star = \boldsymbol{\varepsilon}_{\text{prior}} + \left(\mathsf{A}^\top\mathsf{A} + \lambda\mathsf{I}\right)^{-1}\mathsf{A}^\top\left(\boldsymbol{y} - \mathsf{A}\,\boldsymbol{\varepsilon}_{\text{prior}}\right), \qquad \lambda = \sigma_d^2/\sigma_p^2. \tag{10}$$

The resolution matrix $\mathsf{R} = (\mathsf{A}^\top\mathsf{A} + \lambda\mathsf{I})^{-1}\mathsf{A}^\top\mathsf{A}$ quantifies, per component, the fraction supplied by the data (its diagonal): 1 means measurement-determined, 0 means taken from the prior. On a synthetic build-height profile with a deliberately biased prior, the full set reduces the strain RMSE from 9.4×10^{-4} (prior only) to 2.6×10^{-4}, while the in-plane rosette reaches 3.9×10^{-4}: it corrects the in-plane components from data and falls back to the prior for ε_{zz} and the out-of-plane shears, exactly as the resolution diagonal predicts (Fig. 6). Where only a normal-coil scalar is available, the estimate degrades gracefully to the volumetric strain via the invariant coupling $\kappa_v = (\kappa_\parallel + 2\kappa_\perp)/3$. The scalar λ here assumes an i.i.d. prior and equal probe noise; a full prior covariance estimated from an ensemble of process simulations, together with an instrument-referenced impedance-domain noise model, are natural refinements.

7 Integrated pipeline

The forward model, observability, probe design and inversion are combined in a single open pipeline that ingests layer-wise ECT data and outputs both the reconstructed geometry (for

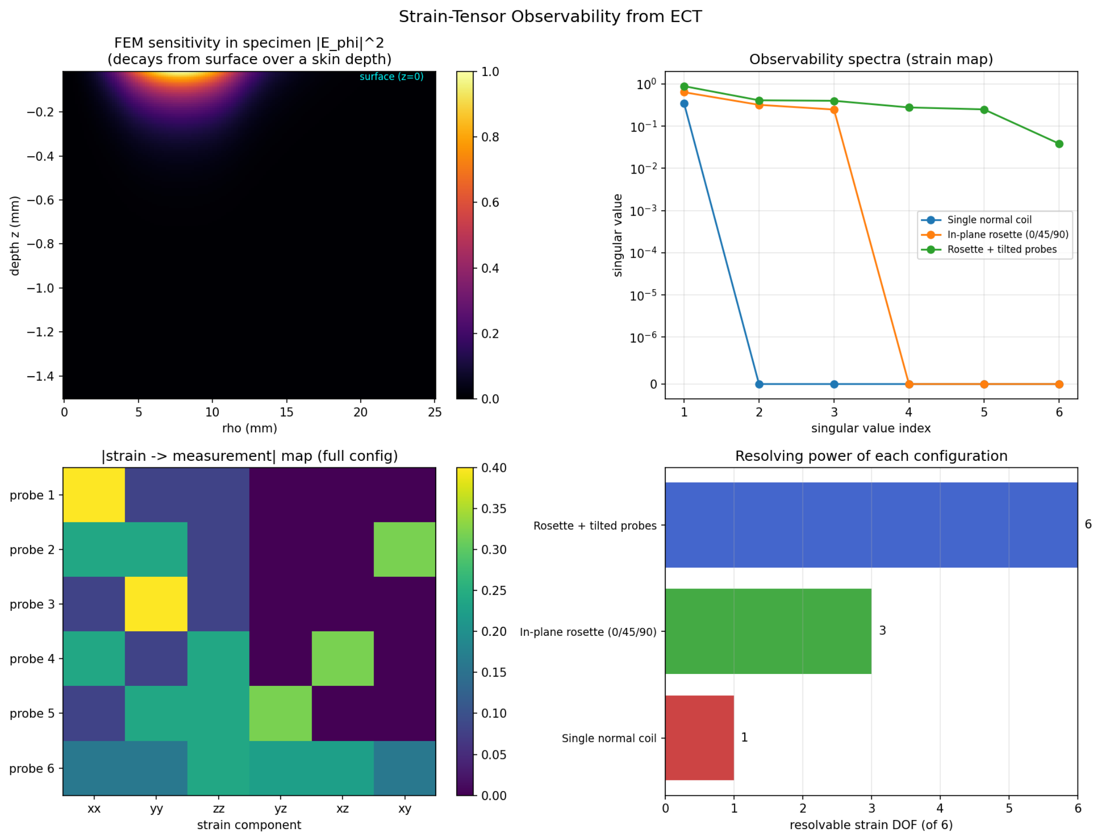

Figure 3: Observability. Top-left: FEM sensitivity density $|E_\phi|^2$ in the specimen, decaying from the surface over a skin depth. Remaining panels: SVD spectra, strain-to-measurement map, and resolving power of the normal coil (1 DOF), in-plane rosette (3 DOF) and full tilted set (6 DOF).

finite-cell simulation) and, from directional measurements, a full strain-*tensor* field. Figure 7 shows a reconstructed $60 \times 40 \times 6$ field for a synthetic build, with the derived volumetric and von-Mises-equivalent strain fields and the per-component resolution. Each voxel is currently inverted independently; the finite spatial footprint of the sensitivity kernel (Fig. 3) couples neighbouring voxels and should, in a quantitative treatment, be folded into A rather than ignored. The implementation is verified by 66 automated tests.

8 Feasibility: signal, confounders and resolution

The framework above is geometry- and material-agnostic; deploying it on LPBF 316L meets three physical realities that set the measurement requirements.

Signal magnitude. Residual elastic strains in LPBF 316L are of order $1\text{–}2 \times 10^{-3}$, so with $|\kappa| \approx 0.33$ the relative conductivity signal is only $\Delta\sigma/\sigma_0 \approx 3\text{–}7 \times 10^{-4}$.

Temperature — the dominant confounder. 316L has resistivity $\rho \approx 0.74\ \mu\Omega\,\mathrm{m}$ ($\sigma \approx 1.35$ MS/m) with a temperature coefficient of order $10^{-3}\,\mathrm{K}^{-1}$ (several $\times 10^{-4}\,\mathrm{K}^{-1}$ near room temperature) [14]. A drift of only $\sim$1 K therefore changes σ by as much as the entire strain signal — consistent with in-situ ECT studies that found part temperature dominant and required explicit compensation [10]. Strain sensing thus demands differential or reference-channel probes, multi-frequency lift-off/temperature separation, and tight thermal referencing. Cold work and

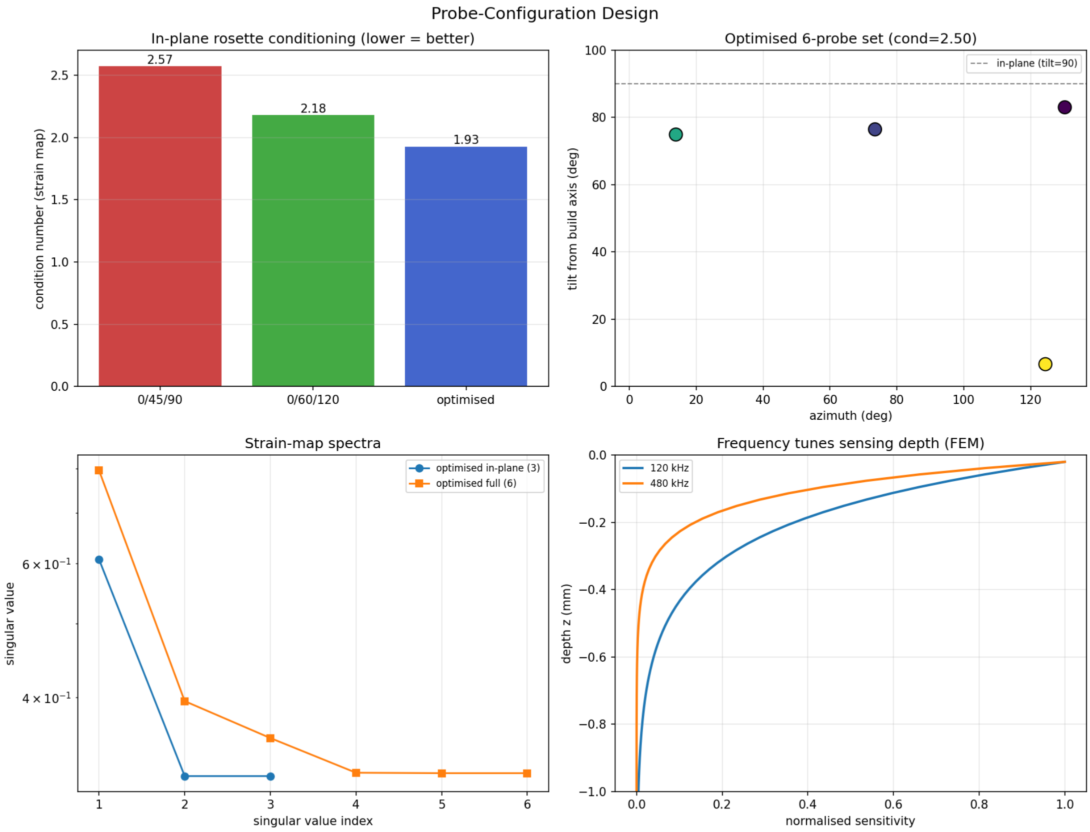

Figure 4: Probe design. In-plane rosette conditioning (top-left), the optimised six-probe set in azimuth/tilt (top-right), strain-map spectra (bottom-left), and the frequency tuning of sensing depth from the FEM model (bottom-right).

dislocation density (Section 1) are a further, microstructure-borne confounder that co-varies with the thermal history.

Penetration and depth resolution. At the realistic 316L conductivity the skin depth is $\delta = \sqrt{2/(\omega\mu\sigma)} \approx 0.9$ mm at 240 kHz (falling to $\approx$0.4 mm at 1 MHz), far larger than a 30–50 μm layer: each measurement averages roughly 20–30 deposited layers. The $60 \times 40 \times 6$ reconstruction of Section 7 is therefore a coarse-z illustration; layer-resolved tomography would require multi-frequency depth inversion, and the forward-model benchmark of Table 1 (a higher, generic conductivity, $\delta \approx 0.25$ mm) should be read as a verification case rather than a 316L prediction.

These constraints do not undermine the framework — they specify the instrument: a thermally-referenced, multi-frequency directional-probe rosette, whose observability and conditioning are exactly what Sections 4–5 quantify.

9 Discussion

The cross-validation between the analytical and FEM forward models proved unexpectedly valuable, exposing two independent errors in a widely used analytical formulation (a reflection-coefficient sign and a units error in the propagation parameter) that a self-consistent synthetic round trip would never reveal; after correction the two models agree to within 1% in resistance and 3% in reactance. This underlines the importance of dissimilar, redundant forward models when an inverse method is to be trusted on real data.

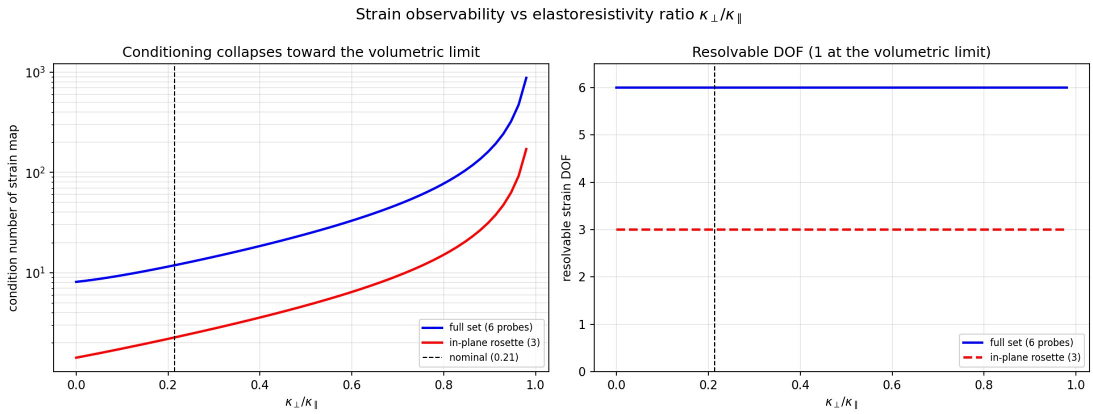

Figure 5: Strain observability vs the elastoresistivity ratio $\kappa_\perp/\kappa_\parallel$. Left: condition number of the strain map for an in-plane rosette and a full set, diverging as the response becomes volumetric. Right: resolvable DOF, full for any anisotropic response. The dashed line marks the thesis-anchored ratio (≈ 0.21).

Two limitations frame the path to experiment. First, surface ECT accesses in-plane current paths much more readily than out-of-plane ones, so ε_{zz} and the out-of-plane shears will in practice lean on the simulation prior or on multi-frequency depth profiling; edge access or tilted probes mitigate this. Second, the tensor model needs calibration: multi-orientation tensile tests are required to populate $\kappa_\parallel, \kappa_\perp$ (and the as-built anisotropy of σ_0 in columnar LPBF microstructures).

10 Conclusion

We have shown how to progress from a scalar strain–conductivity relation to a tensorial reconstruction with ECT. The key realisation is that the scalar impedance constrains a *projection* of the conductivity tensor; resolving the strain tensor then requires measurement diversity (a directional/tilted probe "rosette", optionally multi-frequency) and, for the components that remain unobservable, a physics-based prior fused by regularised inversion. The framework is fully implemented and verified, and — as a by-product of cross-validating two dissimilar forward models — caught two implementation pitfalls in our realisation of the classical model (a reflection sign and a units error in the propagation parameter), after which the two models agree to 0.9% in resistance. Future work targets a 3-D anisotropic eddy-current model to quantify the true out-of-plane sensitivity of tilted and elongated probes, experimental calibration of the elastoresistivity constants, and deployment of the pipeline for in-situ monitoring.

Code and data availability

The open-source pipeline (forward models, observability, probe design and inversion) and the scripts that regenerate every figure are available at `https://github.com/MassimoCarraturo/eddy_current_project`; an archived release with a DOI will be deposited on Zenodo.

CRediT author statement

M. Carraturo: conceptualisation, methodology, software, writing — original draft. **B. I. Ates:** calibration data, methodology, writing — review and editing.

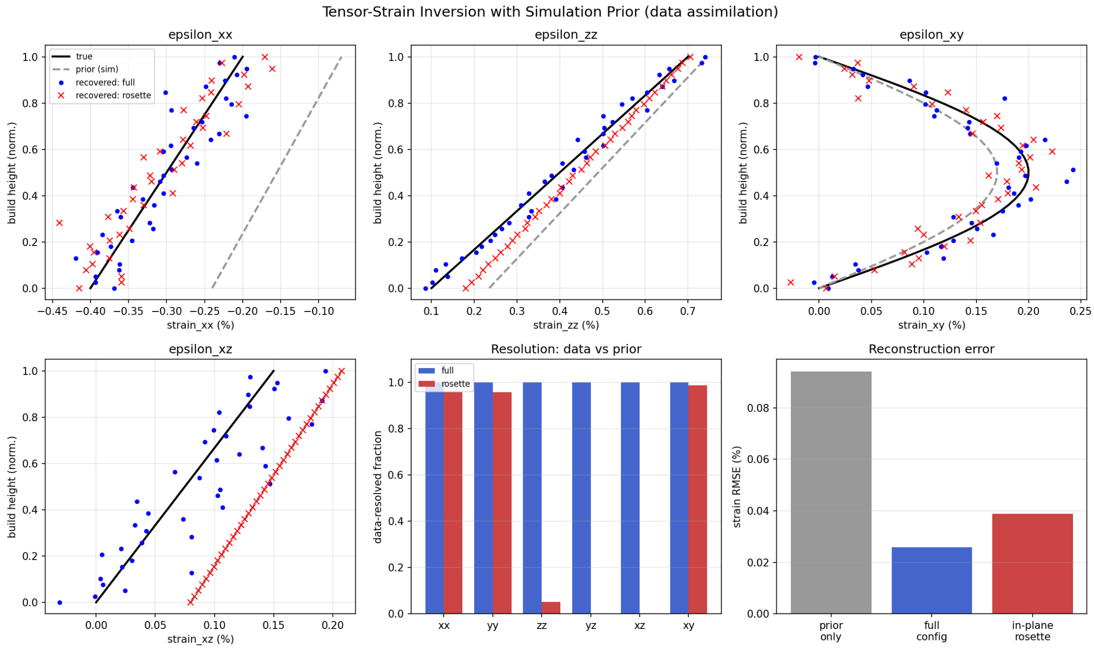

Figure 6: Tensor-strain inversion with a biased simulation prior. In-plane components ($\varepsilon_{xx}, \varepsilon_{xy}$) are corrected from data by both configurations; out-of-plane components ($\varepsilon_{zz}, \varepsilon_{xz}$) are recovered only by the full set and otherwise follow the prior, consistent with the resolution diagonal (centre).

Declaration of generative AI

Generative-AI tools assisted software development and manuscript editing; all scientific content was verified by the authors, who take full responsibility for it. AI tools are not listed as authors.

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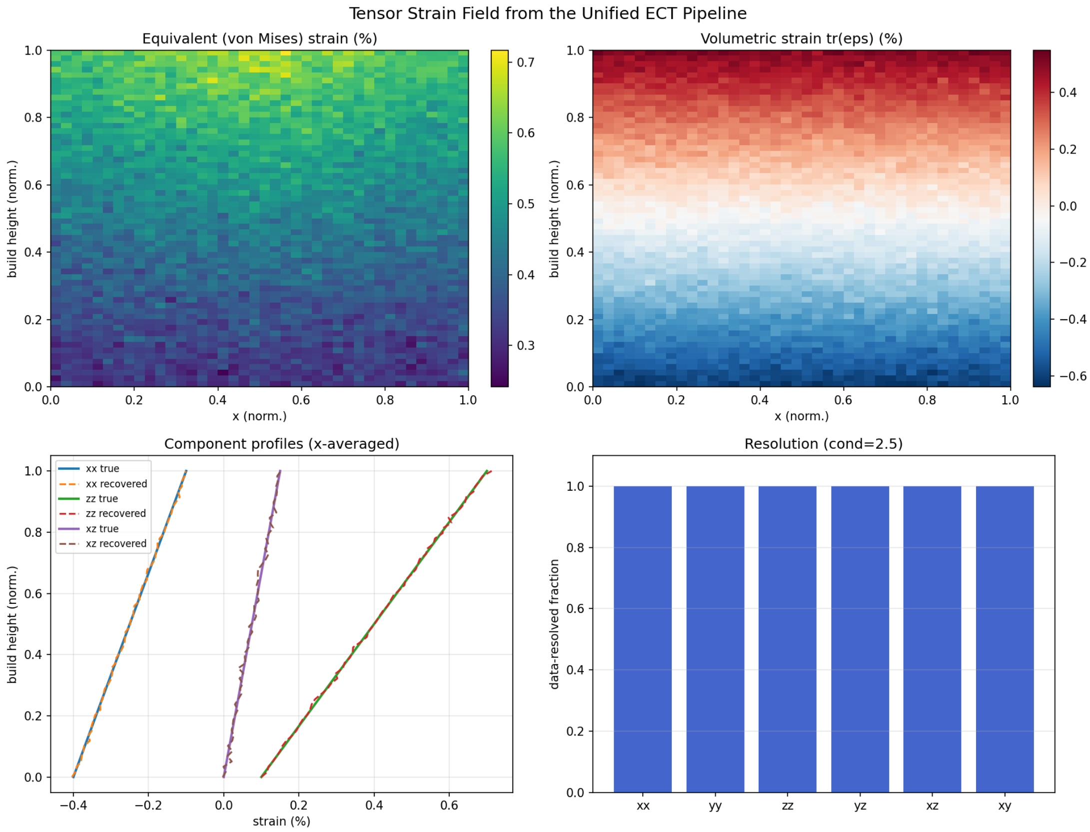

Figure 7: Strain-tensor field reconstructed by the unified pipeline. Equivalent (von Mises) and volumetric strain fields over the build (top), component profiles versus build height (bottom-left) and the per-component data resolution (bottom-right).

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